Hybrid Networks Overview

Course: Networking Principles in Practice – Cloud Networking
Module: Hybrid



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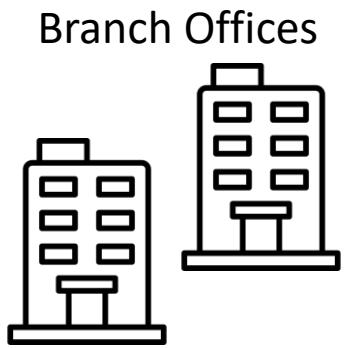


We’ve focused on Google Cloud,
where all of your infrastructure is GCP



Companies will often have some IT infrastructure in their corporate offices that needs to interface to the infrastructure deployed in the cloud.

e.g., data is stored in GCP, but employees need to be able to access it and make queries on it from the office





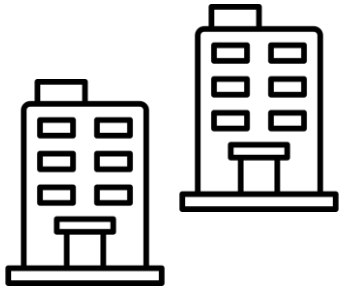
Sometimes the infrastructure is hosted in co-location facilities.

e.g., company built their infrastructure around co-lo, and wants to expand into the cloud

Co-location Facility



Branch Offices









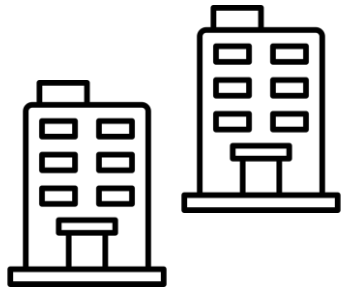
Maybe a company wants to use multiple clouds

e.g., they want access to the OpenAI infrastructure in Azure

Co-location Facility



Branch Offices



Network Connectivity

- All of these need connectivity
- Trading off different properties:
  - Security
  - Reliability
  - Performance (Latency/throughput)
  - Cost
  - Development effort

Module Outline

- VPN (from branch offices, or really anywhere)
- Direct Connections (from co-location)
- Inter-cloud connections



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VPN

Course: Networking Principles in Practice – Cloud Networking
Module: Hybrid

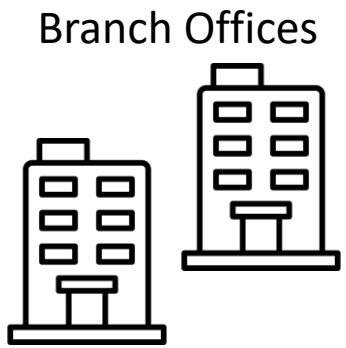


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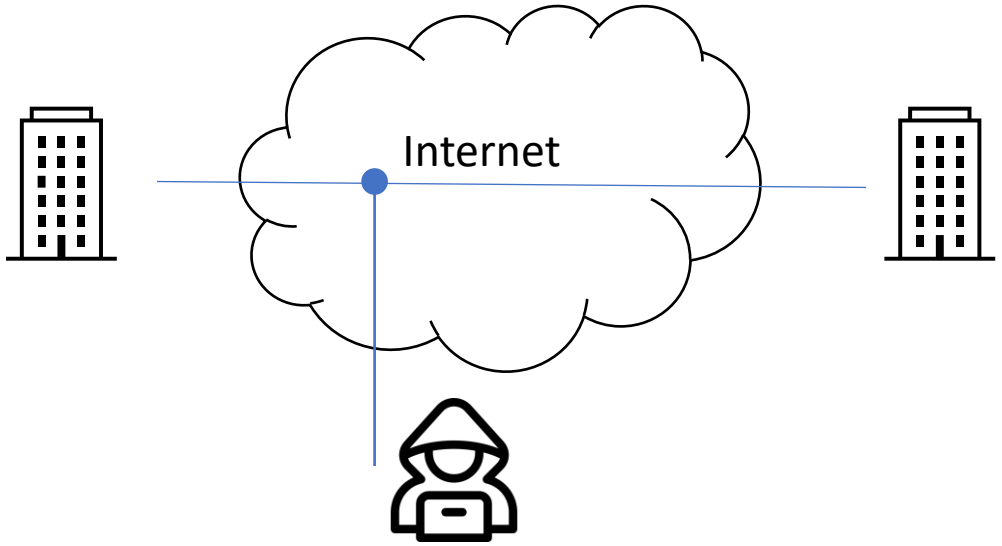
Companies will often have some IT infrastructure in their corporate offices that needs to interface to the infrastructure deployed in the cloud.

e.g., data is stored in GCP, but employees need to be able to access it and make queries on it from the office



Why VPN is needed?

- Company has 2 sites and employees at different sites want to share information with each other (or data in cloud employee need to access)
- If attacker can see the traffic, will get access to secret data.
- Soln: IPsec

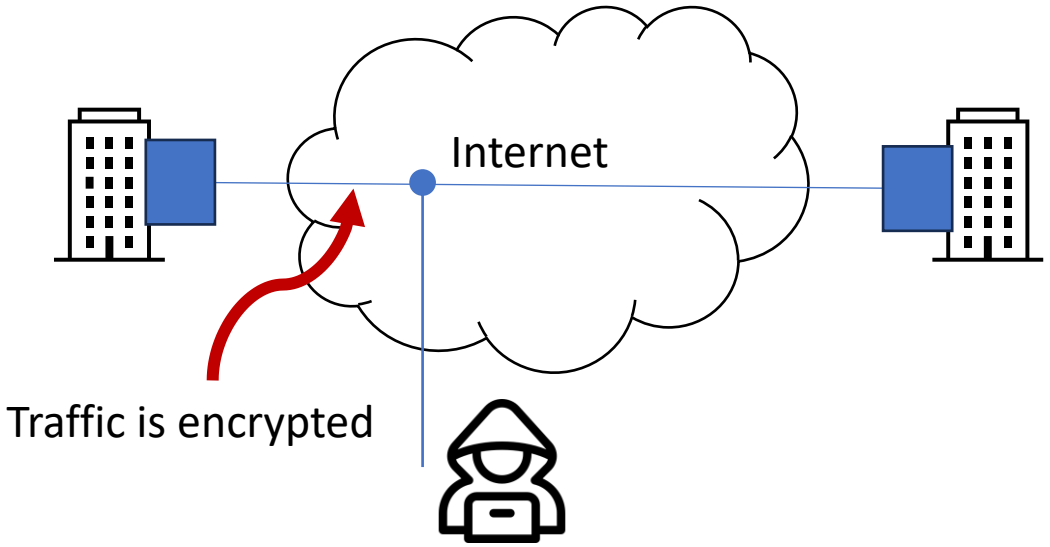


How Would Someone Eavesdrop?

- Wi-fi is broadcast communication
- Network provider compelled by government agency or has a rogue employee or a compromised router
- Misconfiguration in routing redirects traffic to the wrong place
- Route hijacking to intentionally redirect traffic

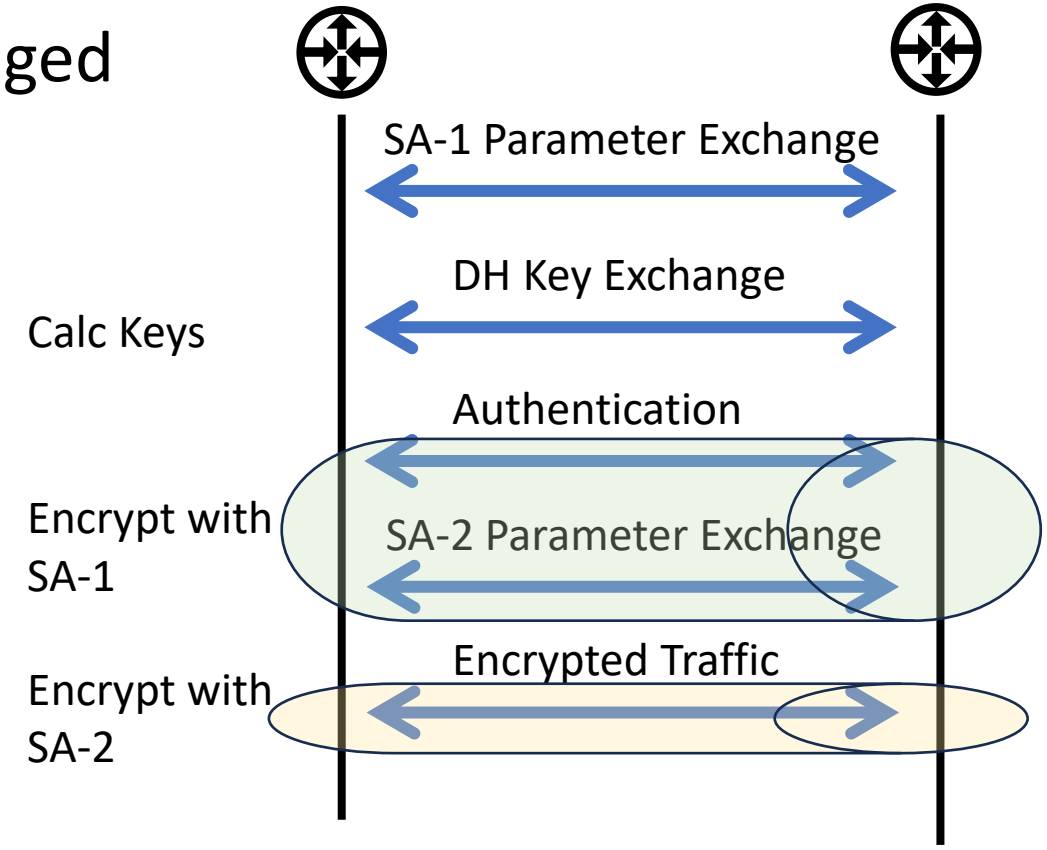
IPsec

- Create an encrypted tunnel between two end points
  - Can be for whole site or one machine
- Commonly used in VPNs (virtual private networks)

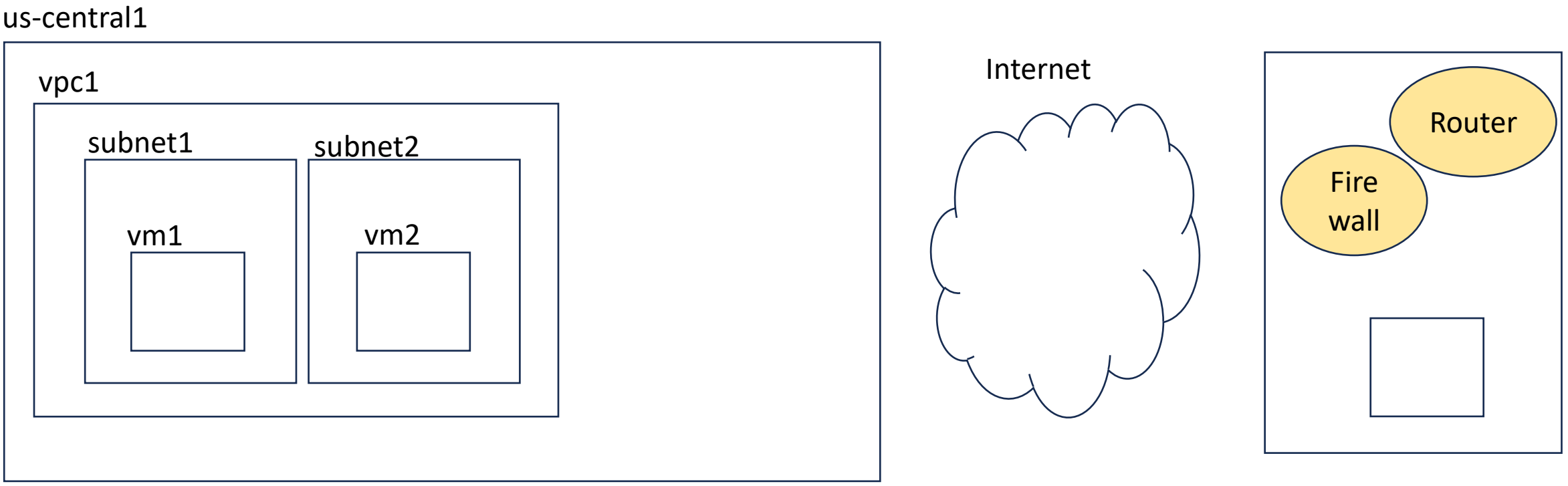


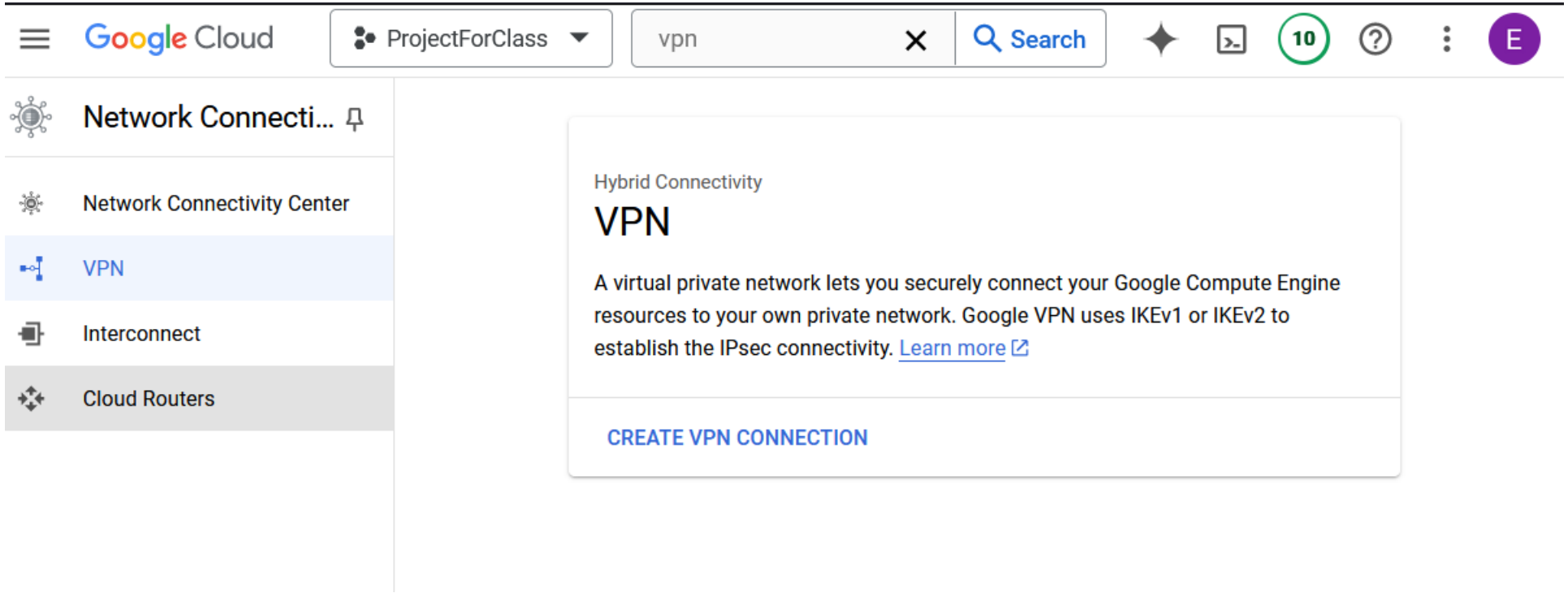
IPsec Overview

- To set up an IPsec tunnel, a protocol (ISAKMP/IKE) is used to set up the parameters and keys used, and authenticate each side
- Then traffic can be exchanged

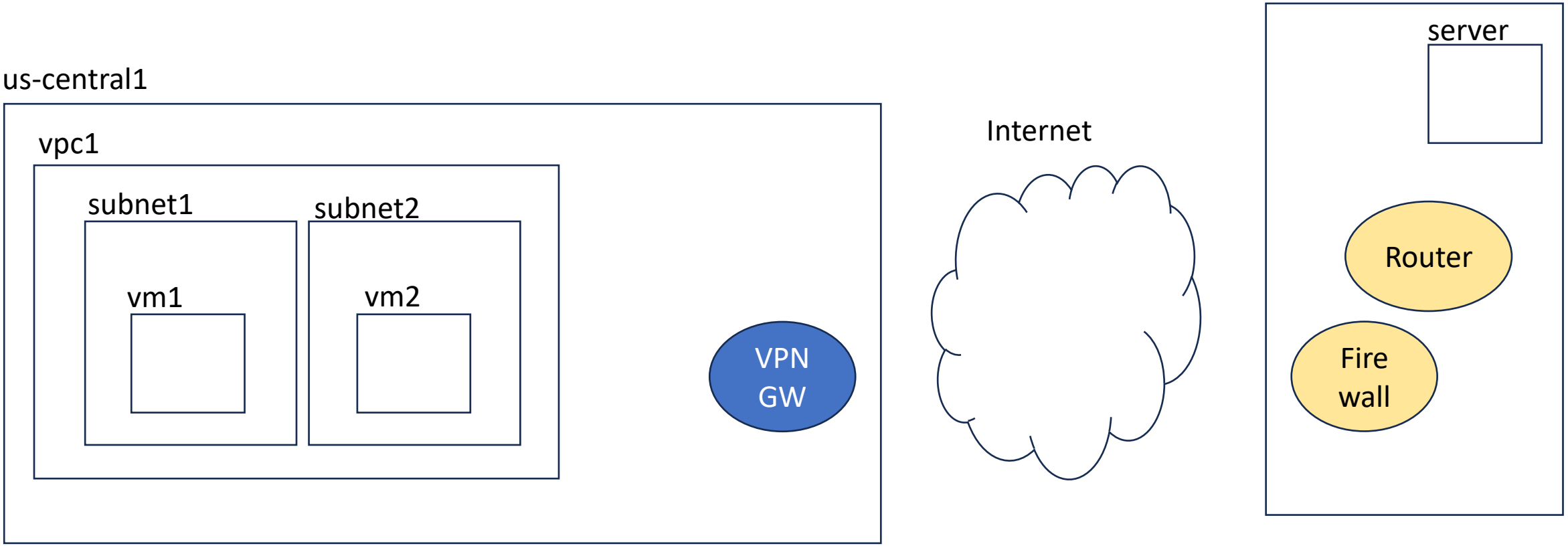


ISAKMP - Internet Security Association and Key Management Protocol
IKE – Internet Key Exchange
SA – Security Association


us-central1
vpc1
subnet1
subnet2
vm1
vm2
Internet
Router
Fire
wall


Google Cloud
ProjectForClass
vpn
Search
10
E
Network Connecti...
Network Connectivity Center
VPN
Interconnect
Cloud Routers
Hybrid Connectivity
VPN
A virtual private network lets you securely connect your Google Compute Engine resources to your own private network. Google VPN uses IKEv1 or IKEv2 to establish the IPsec connectivity. Learn more
CREATE VPN CONNECTION

Add a VPN Gateway

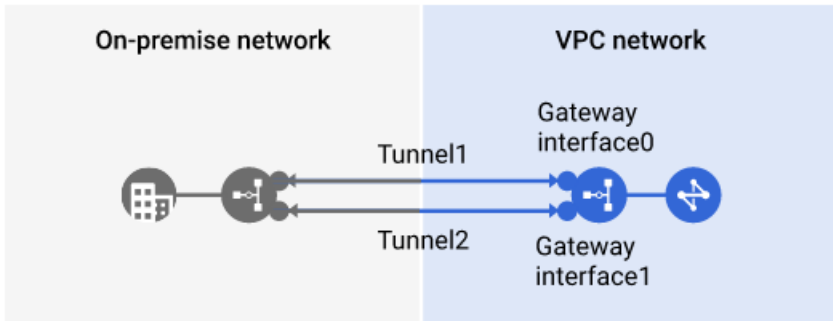


Create a VPN

A virtual private network lets you securely connect your Google Compute Engine resources to your own private network. Google VPN uses IKEv1 or IKEv2 to establish the IPsec connectivity. Learn more

VPN options

- (•) High-availability (HA) VPN
  Supports dynamic routing (BGP) only
  Supports high availability (99.99 SLA, within region)
  Supports IPv4 and IPv6 traffic
  Learn more



- ( ) Classic VPN
  Supports static routing
  No high availability
  Supports IPv4 traffic only
  Learn more



1. Create Cloud HA VPN gateway
2. Add VPN tunnels
3. Configure BGP sessions
4. Summary and reminder

Create Cloud HA VPN gateway

High Availability (HA) capable Cloud VPN gateways are regional resources with two interfaces, each interface with its own external IP address. HA VPN connects to an on-premises VPN gateway or another Cloud VPN gateway. Learn more

VPN gateway name *

Lowercase letters, numbers, hyphens allowed

Network *

Region *

Region is permanent

VPN gateway public IP address

Two IP addresses will be automatically allocated for each of your gateway interfaces

VPN tunnel inner IP stack type

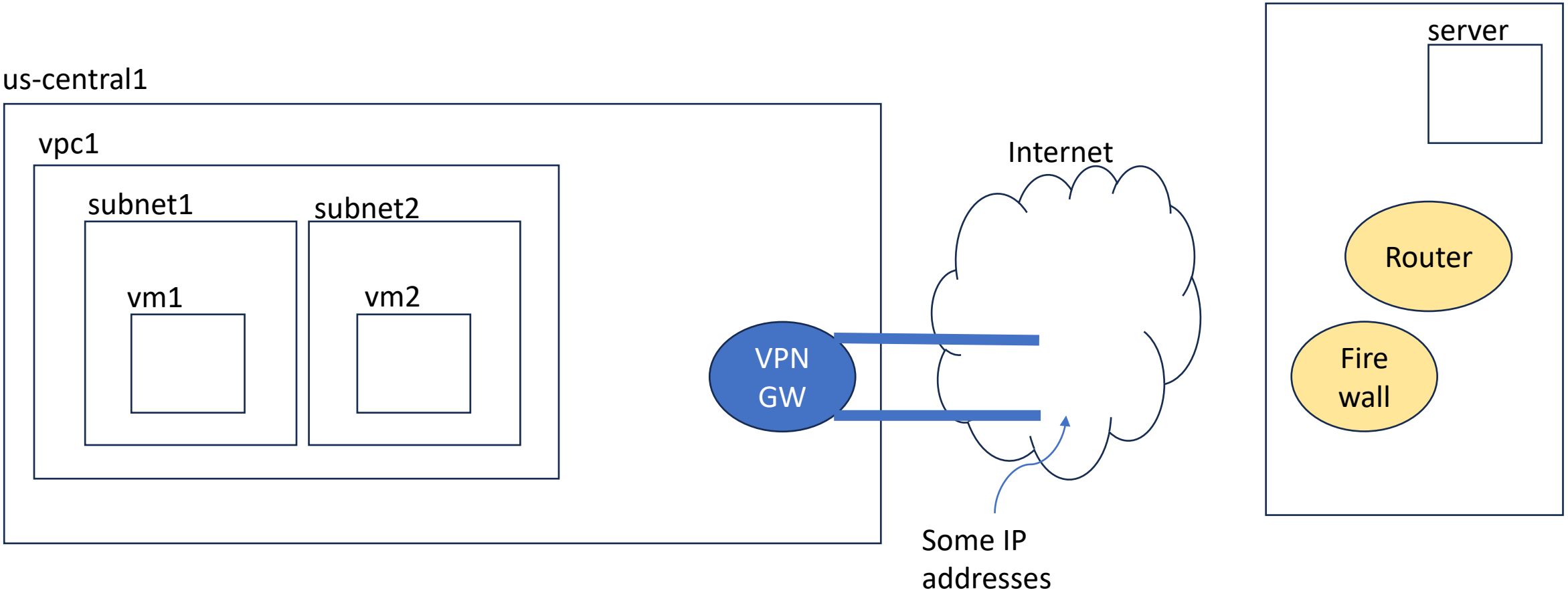
The IP stack type will apply to all the tunnels associated with this VPN gateway.

- (•) IPv4 (single-stack)
- ( ) IPv4 and IPv6 (dual-stack)

CREATE & CONTINUE CANCEL

?project=orbital-linker-398719

Add Tunnels


us-central1
vpc1
subnet1
subnet2
vm1
vm2
VPN
GW
Internet
Some IP
addresses
server
Router
Fire
wall

- ✓ Create Cloud HA VPN gateway
- 2 Add VPN tunnels
- 3 Configure BGP sessions
- 4 Summary and reminder

Add VPN tunnels

A VPN tunnel connects the Cloud VPN gateway to a peer gateway. Traffic sent through the tunnel is encrypted using the IPsec protocol operating in tunnel mode. Learn more

VPC network	testvpc
Region	us-central1
VPN gateway name	vpn-gw
Interfaces	0 : 34.157.97.117 1 : 35.220.70.106

Peer VPN gateway

- (•) On-prem or Non Google Cloud
- ( ) Google Cloud VPN Gateway
- ( ) Compute Engine VMs with external IP addresses

Peer VPN gateway name *

Peer VPN gateway is required

You can add more VPN tunnels to the same VPN gateway afterwards

CREATE & CONTINUE CANCEL

Add a peer VPN gateway

This peer VPN gateway resource represents your remote peer gateway. The peer VPN gateway can be an on-premises gateway, a third-party VPN service, or an external IP address of a Compute Engine VM. Learn more

Name *

Lowercase letters, numbers, hyphens allowed

Peer VPN gateway interfaces

Interfaces

- ○ one interface
- ◉ two interfaces
- ○ four interfaces

Interface 0 IP address *

Interface 1 IP address *

CREATE CANCEL

1. Create Cloud HA VPN gateway
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4. Summary and reminder

Routing options

Dynamic (BGP)

Cloud Router *

Turn on global dynamic routing for network 'testvpc' to allow this router to dynamically learn routes to and from all Google Cloud regions on a network. If you're using an internal load balancer with VPN or Interconnect, learn how global dynamic routing may affect you.

Create a router

Google Cloud Router dynamically exchanges routes between your Virtual Private Cloud (VPC) and on-premises networks by using Border Gateway Protocol (BGP)

Name *

Lowercase letters, numbers, hyphens allowed

Description

Network
testvpc

Region
us-central1 (Iowa)

Google ASN *

BGP peer keepalive interval seconds

Advertised routes

Routes

- (•) Advertise all subnets visible to the Cloud Router (Default)
- ( ) Create custom routes

CREATE CANCEL

1. Create Cloud HA VPN gateway
2. Add VPN tunnels
3. Configure BGP sessions
4. Summary and reminder

VPN tunnels

Expand each item to fill in the details of created VPN tunnels

Edit VPN tunnel

Associated Cloud VPN gateway interface
0 : 34.157.97.117

Associated peer VPN gateway interface *
0 : 1.2.3.4

Name *

Lowercase letters, numbers, hyphens allowed

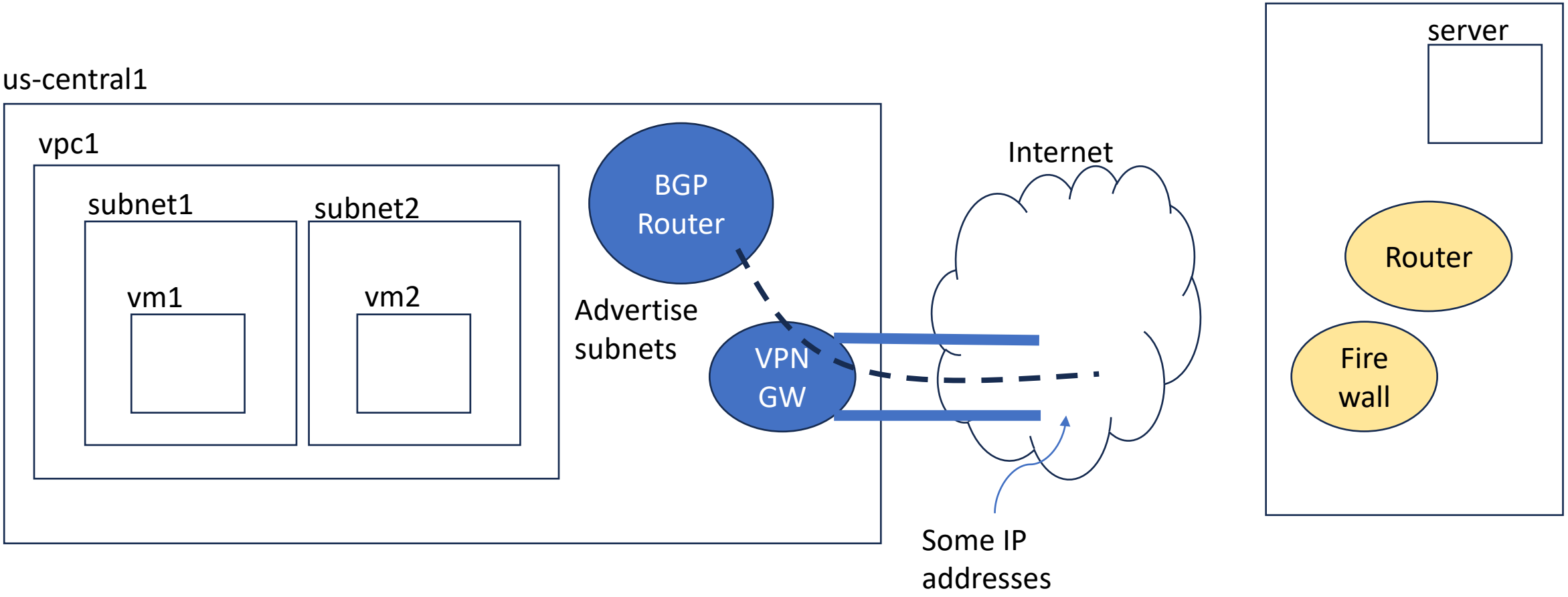
Description

IKE version
IKEv2

IKE pre-shared key * GENERATE AND COPY

Enter your own key or generate one automatically

Add Router and Setup BGP Peering



- ✓ Create Cloud HA VPN gateway
- ✓ Add VPN tunnels
- 3 Configure BGP sessions
- 4 Summary and reminder

Configure BGP sessions

Click Configure BGP Session to set up the BGP session on the Cloud Router cr1 for each tunnel.

BGP session	Cloud VPN tunnel	Inner IP stack type	Cloud VPN gateway	Cloud VPN gateway interface	
(not configured)	tun1	IPv4	vpn-gw	0 : 34.157.97.117	CONFI
(not configured)	tun2	IPv4	vpn-gw	1 : 35.220.70.106	CONFI

SAVE BGP CONFIGURATION CONFIGURE BGP SESSIONS LATER

Create BGP session

Name *

Lowercase letters, numbers, hyphens allowed

Peer ASN *

Advertised route priority (MED)

MED value is used for Active/Passive configuration

Multiprotocol BGP

BGP sessions are set up over IPv4, exchanging IPv4 and IPv6 addresses

Enable IPv6 traffic

You need to enable IPv6 traffic to allocate BGP next hops for IPv6 traffic.

Allocate BGP IPv4 address

- (•) Automatically
- ( ) Manually

BGP peer

- (•) Enabled
- ( ) Disabled

Advertised routes, custom learned routes

Advertised routes

Routes

- (•) Use Cloud Router's advertisements (Default)
- ( ) Create custom routes

Custom learned routes

Priority of all custom learned routes

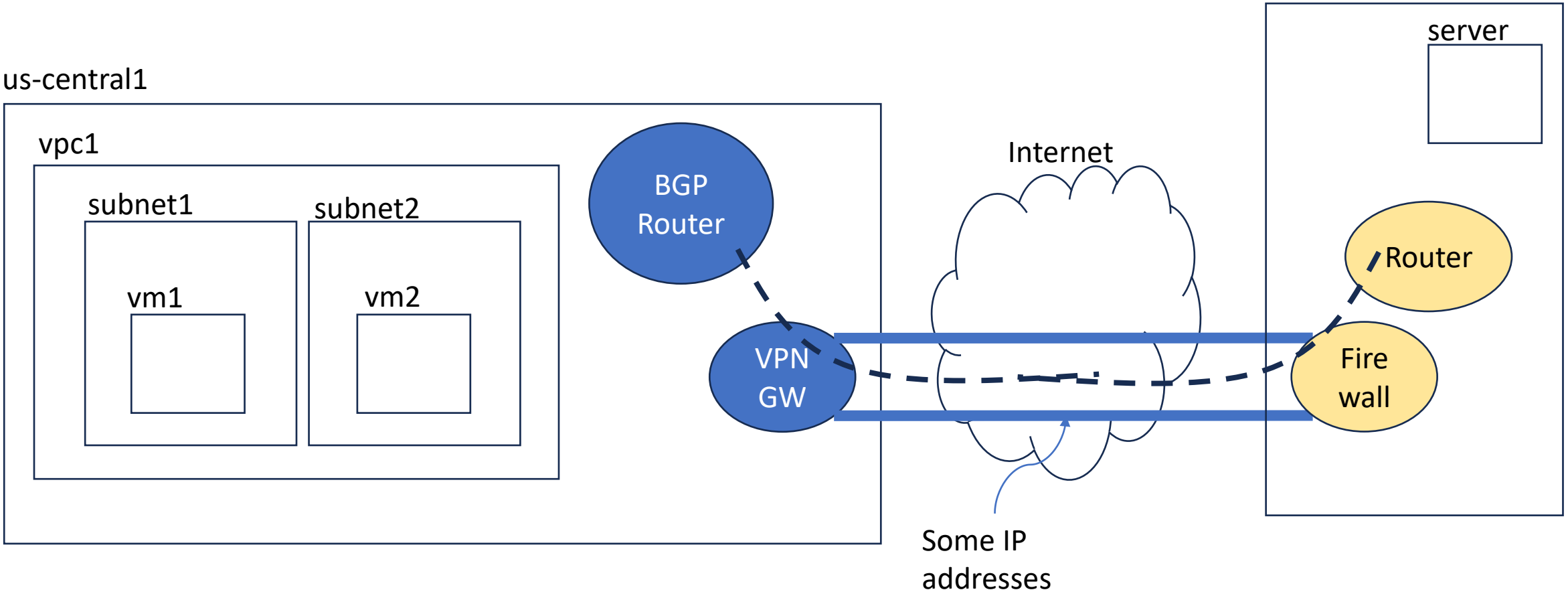
If not specified, a Google-managed priority value 100 will be used.

IP address ranges

+ ADD NEW IP ADDRESS RANGE

SAVE AND CONTINUE CANCEL

Setup Private Side Router and Firewall (VPN)



- ✓ Create Cloud HA VPN gateway
- ✓ Add VPN tunnels
- ✓ Configure BGP sessions
- 4 Summary and reminder

Cloud VPN tunnel(s)

Name	VPN tunnel status	Cloud VPN gateway interface	BGP session	BGP status	Multiprotocol BGP
tun1	⚠ First handshake	0 : 34.157.97.117	bgp1	⚠ Waiting for peer	Disabled IPv6
tun2	⚠ First handshake	1 : 35.220.70.106	bgp2	⚠ Waiting for peer	Disabled IPv6

Peer VPN gateway profile

vpn-gw-peer

Reminder

To finish establishing connections, make sure to complete the following

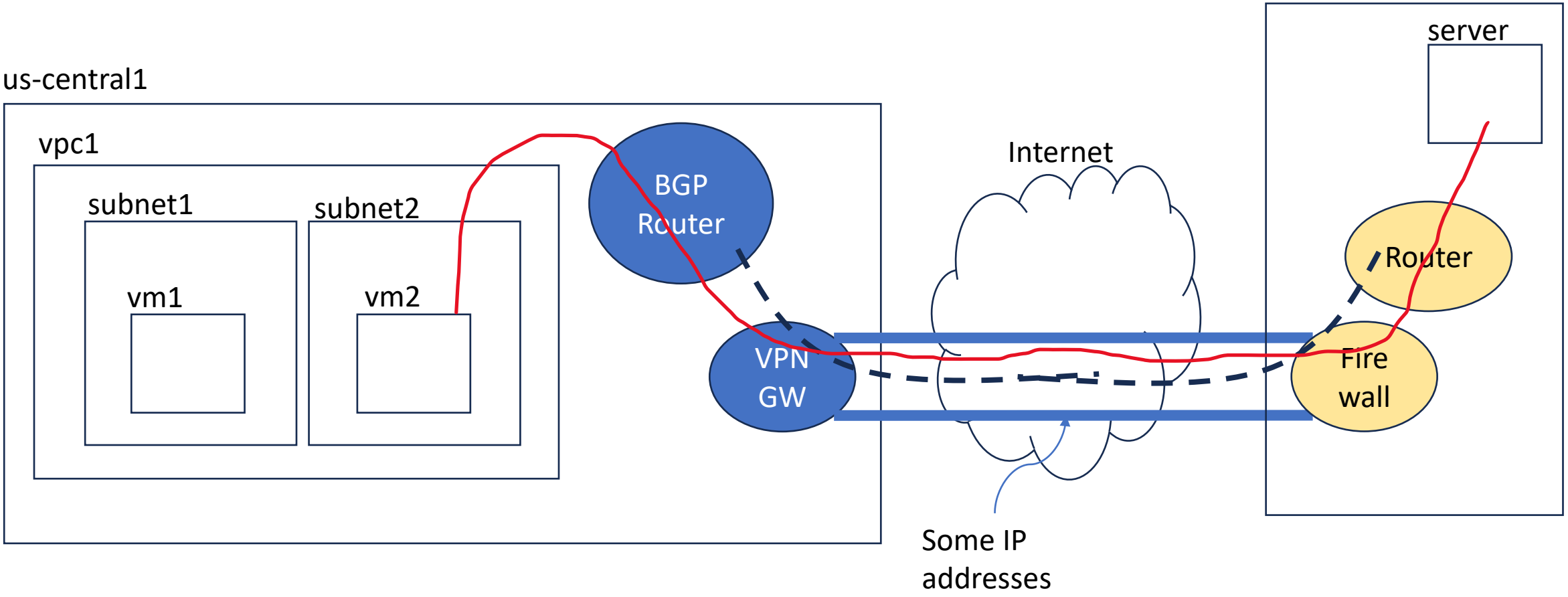
1. To complete your VPN connection, you must configure your peer VPN gateway with information that provides access to your VPC network. Download configuration to start a configuration file for your peer VPN gateway.

DOWNLOAD CONFIGURATION

For additional information about configuring your peer VPN gateway or device, see the documentation

OK

Server and VMs can Communicate Privately



Final Note

- IPsec connection not restricted to an office.
- AWS - https://aws.amazon.com/vpn/
- Azure - https://azure.microsoft.com/en-us/products/vpn-gateway
- Key is to use IP address (as peer) assigned by AWS or Azure



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Direct Connect

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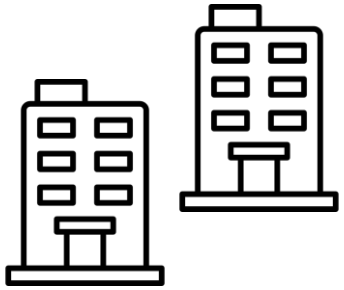
Sometimes the infrastructure is hosted in co-location facilities.

e.g., company built their infrastructure around co-lo, and wants to expand into the cloud

Co-location Facility



Branch Offices



Co-location

- Provider owns/operates a physical building (commonly popular areas)
- Customers can place their physical equipment in the building
- Co-location facility provides:
  - space,
  - power,
  - network connectivity, and
  - physical security







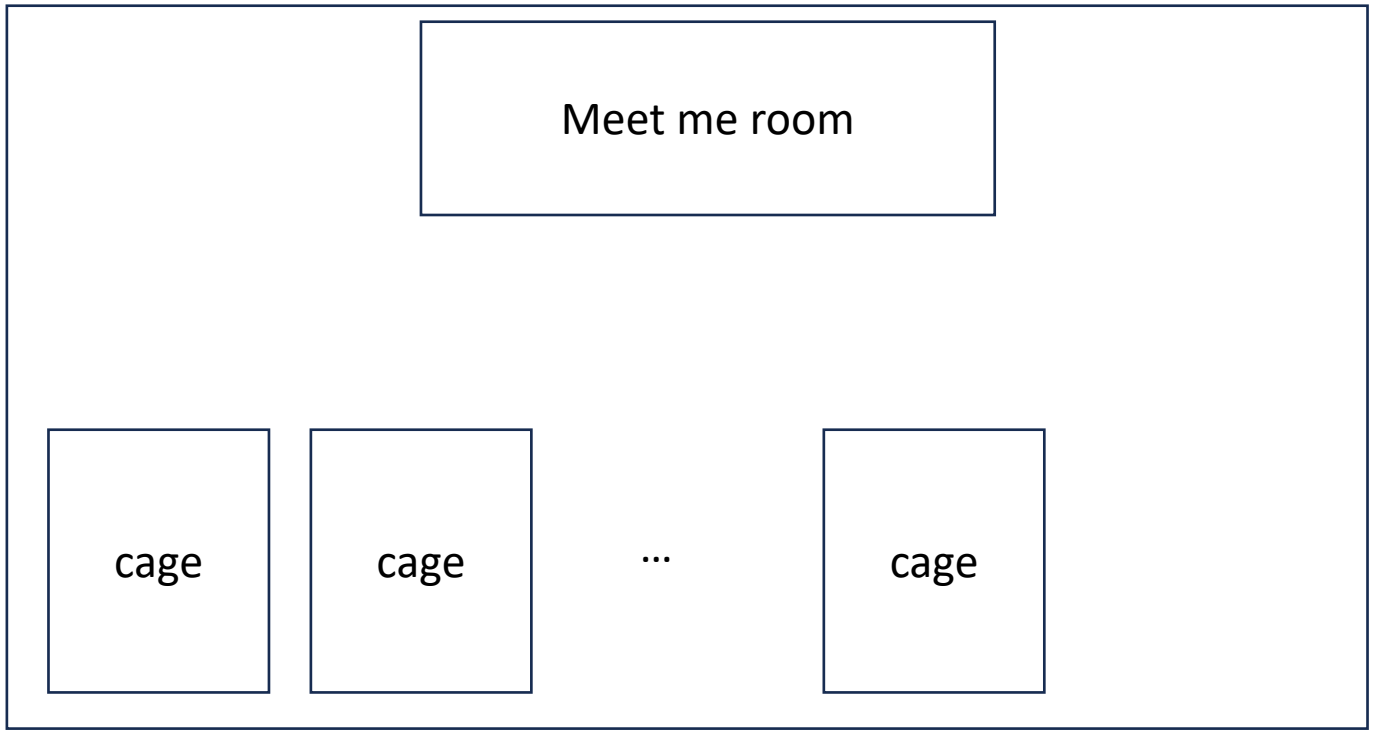
Cage

In addition to building access security, each company gets own secured cage



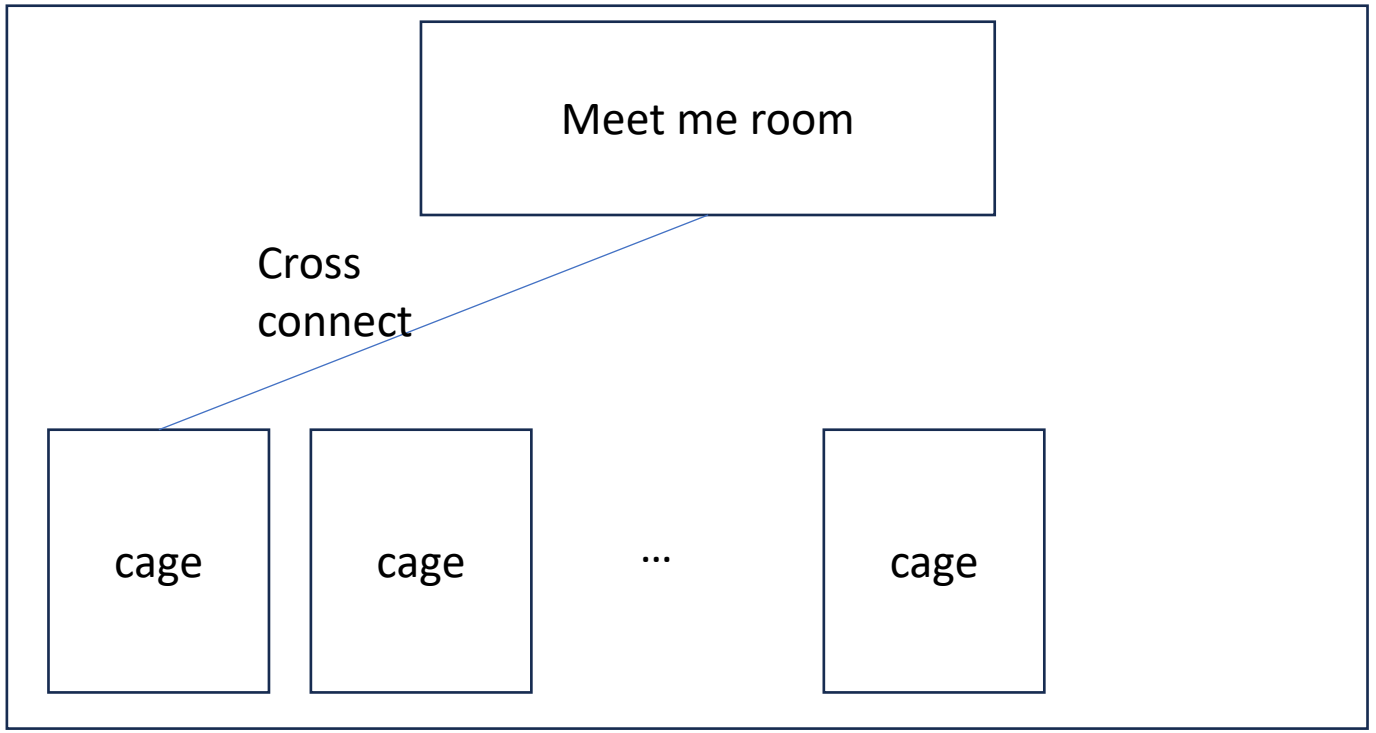
Meet me room

- Network provider equipment is placed in a meet me room



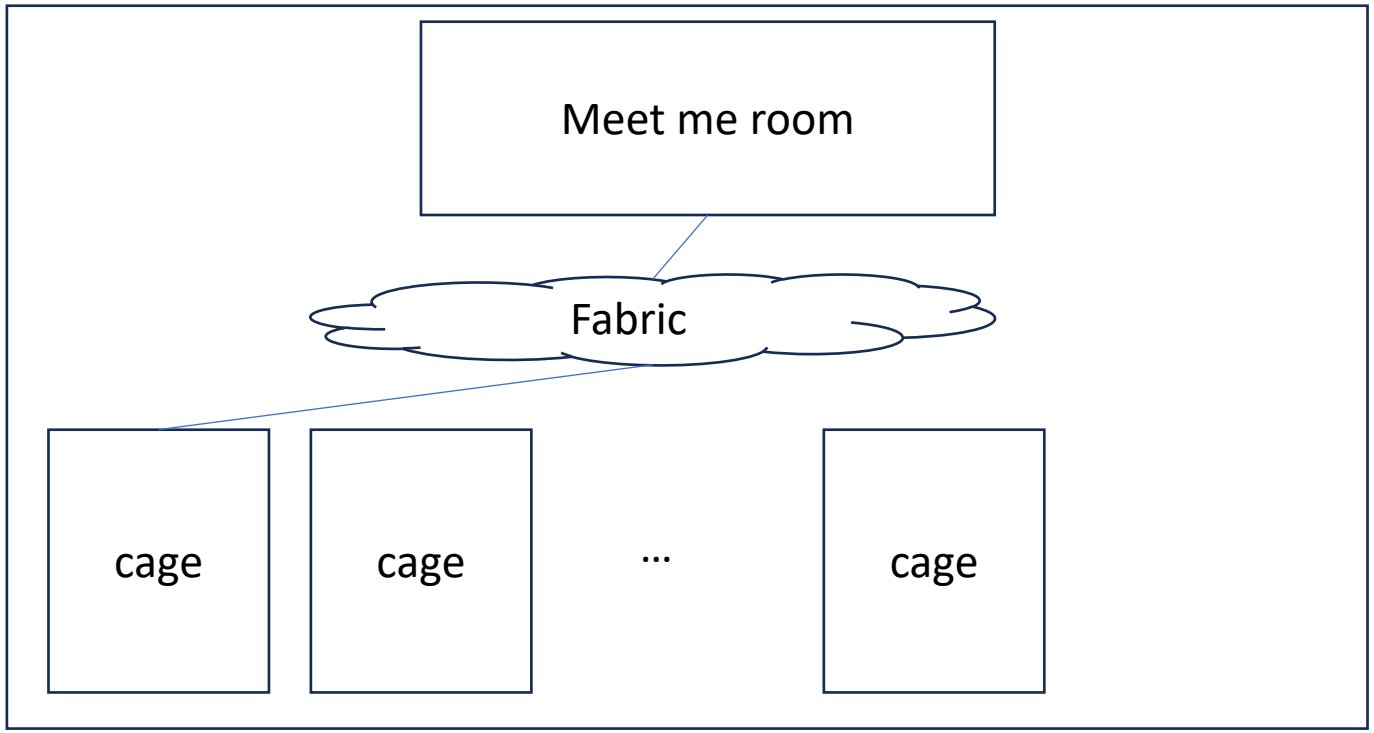
Cross Connects

- Cross connects can connect a cage to network providers



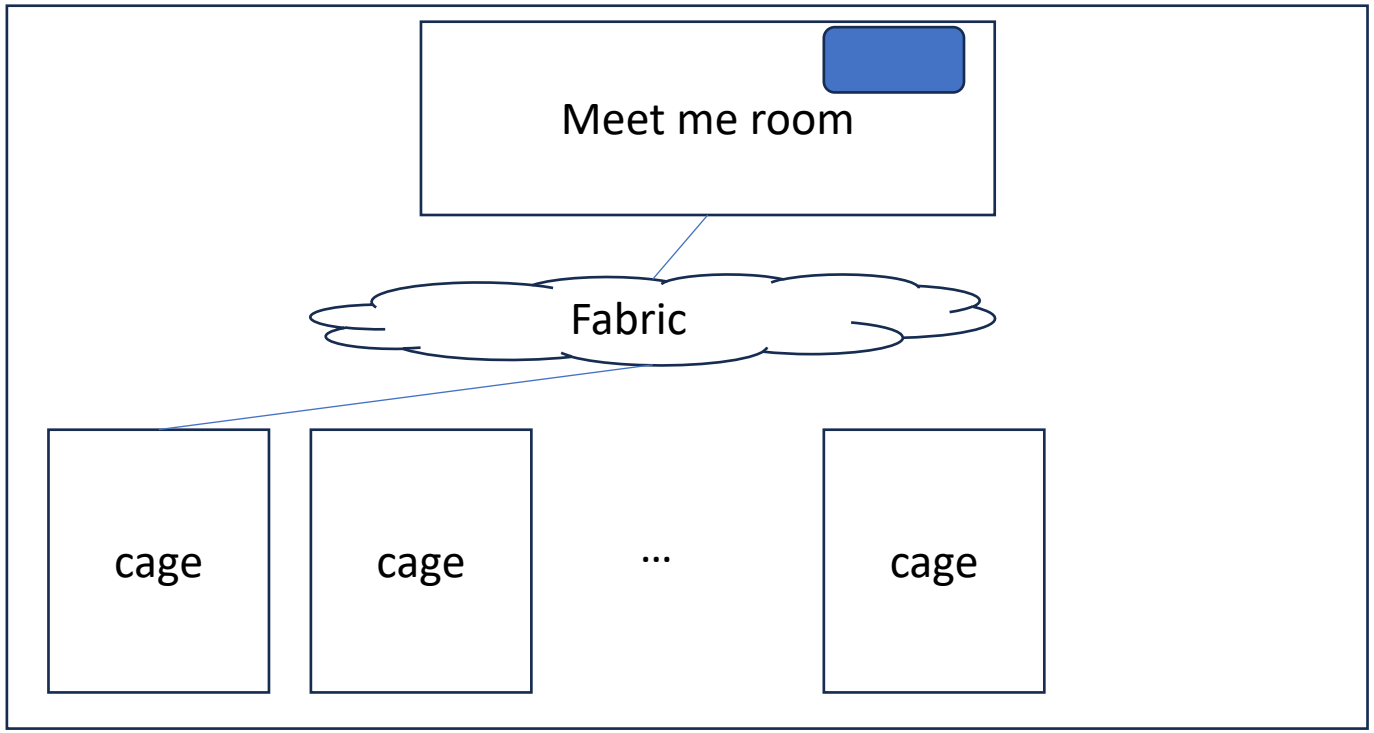
Virtual Cross Connects / Fabric

- Co-lo provider's moving towards a network fabric to help automate cross-connect connections



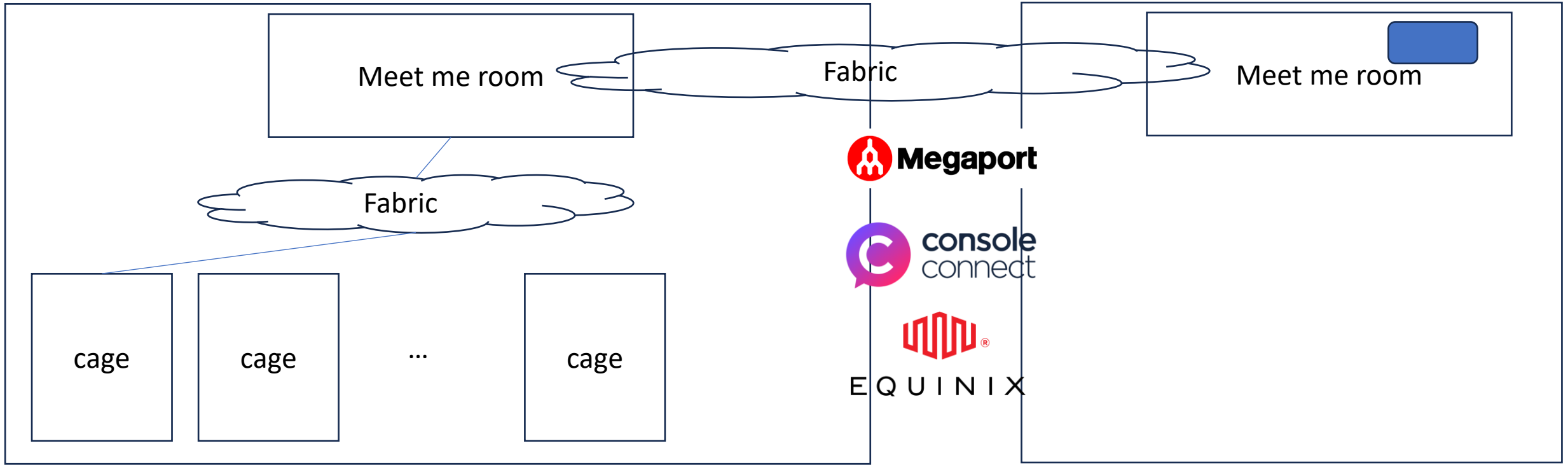
Cloud Direct Connect

- In some facilities, cloud providers have network equipment
- Customers can get a port on, e.g., Google's physical switch which connects them to Google Cloud



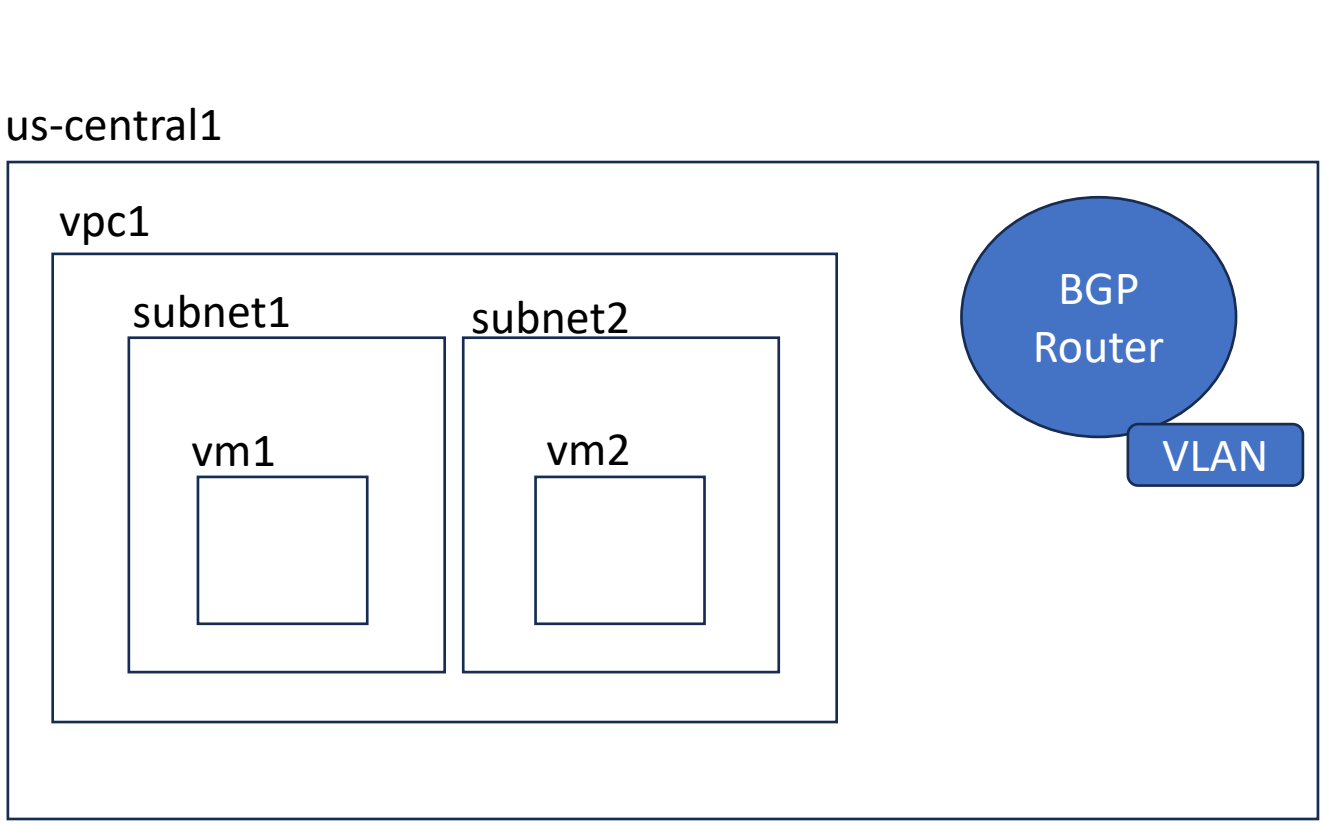
Cloud Direct Connect Providers

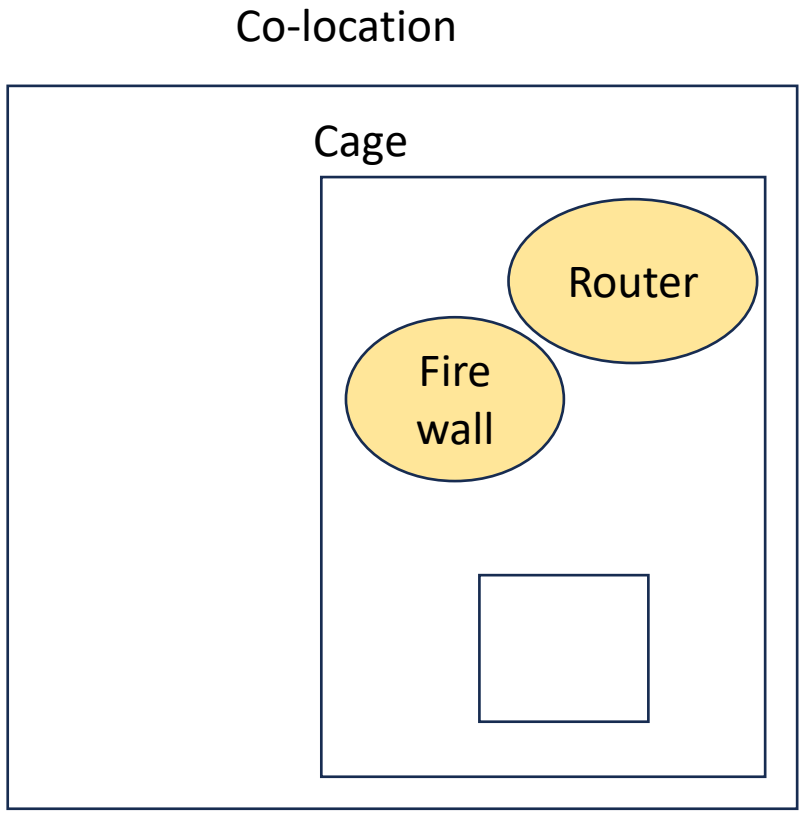
- Some provider's have a global network fabric and can 1) automate some of the connection process, 2) enable connections beyond where the cloud provider switch exists.



Example Setup

- After ordering the cross connect, and direct connect
- In cloud, need Router and VLAN attachment





Network Connecti...

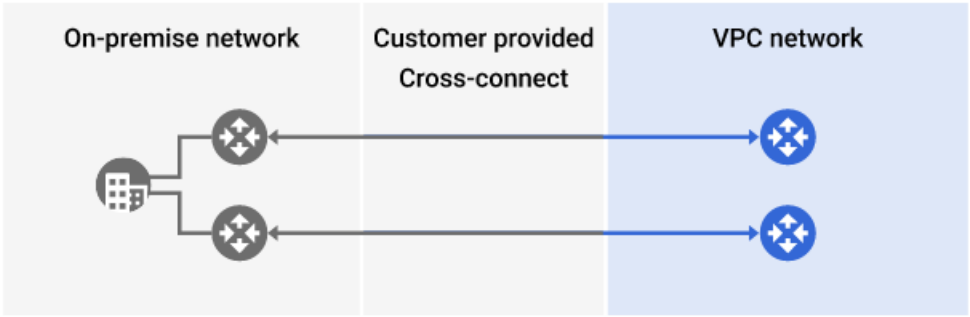
Add VLAN attachment

- Network Connectivity Center
- VPN
- Interconnect
- Cloud Routers

Interconnect type

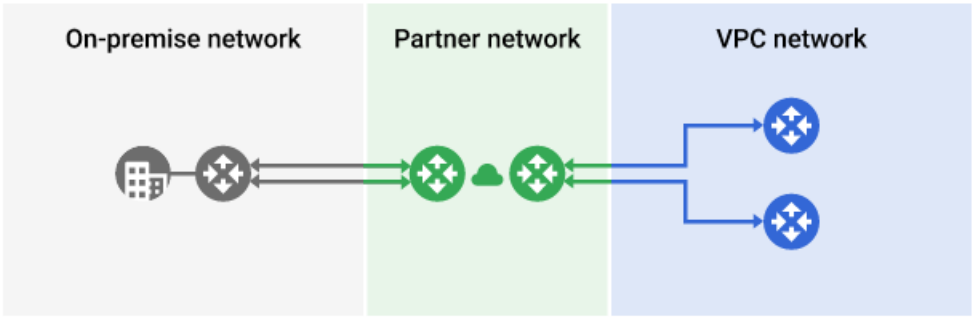
Dedicated Interconnect connection

Connect your on-premises network to your Google Cloud VPC network by connecting a new fiber to your equipment. **Supports IPv4 and IPv6 traffic.** Learn more



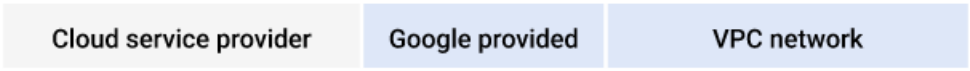
Partner Interconnect connection

Connect your on-premises network to your Google Cloud VPC network through a connection from a supported service provider. **Supports IPv4 traffic only.** Learn more or check supported service providers

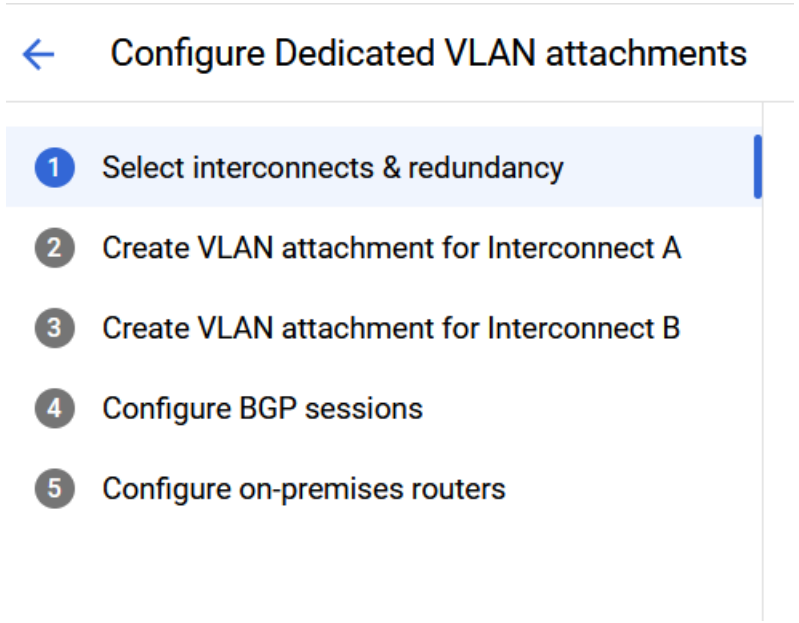


Cross-Cloud Interconnect connection

Connect your Google Cloud VPC network to your other cloud service provider network. Learn more



Another Wizard





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Inter-cloud

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Module: Hybrid



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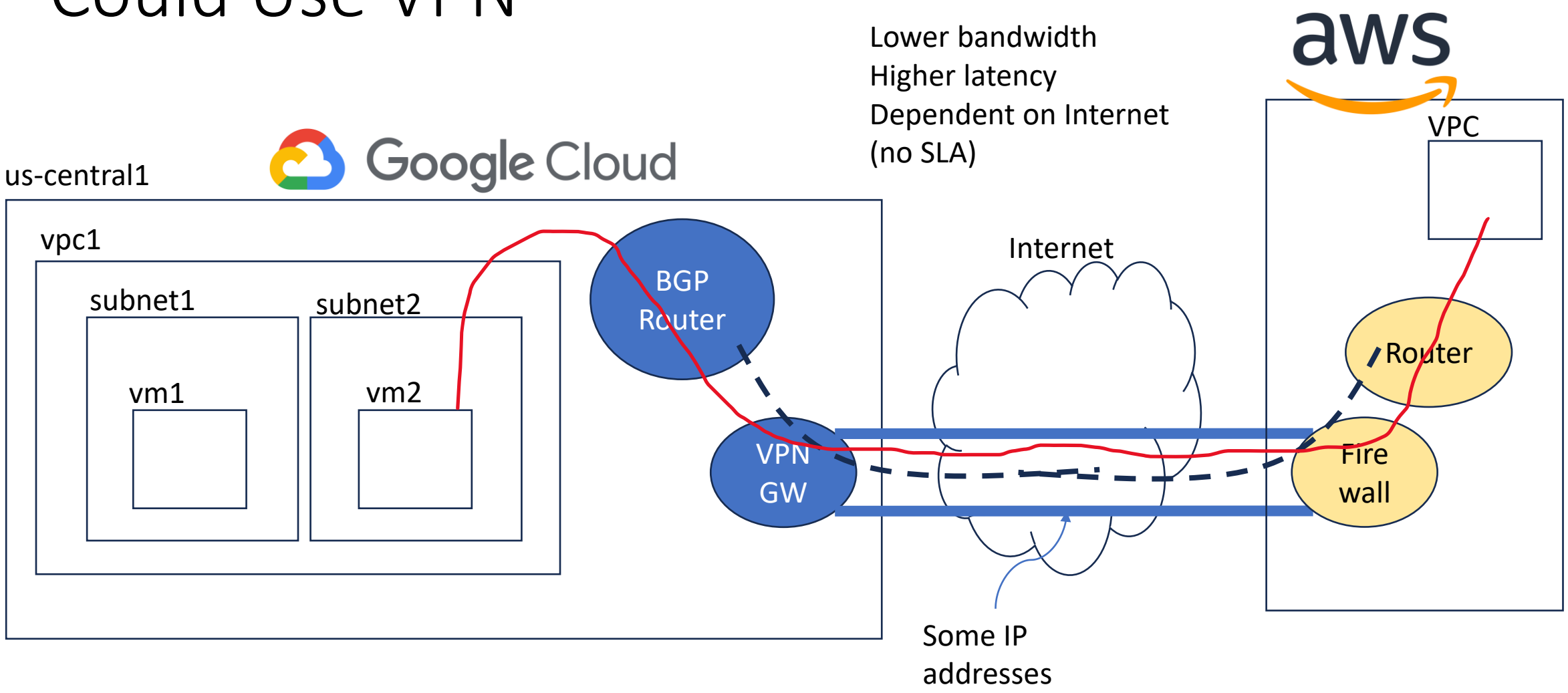


Maybe a company wants to use multiple clouds

e.g., they want access to the OpenAI infrastructure in Azure

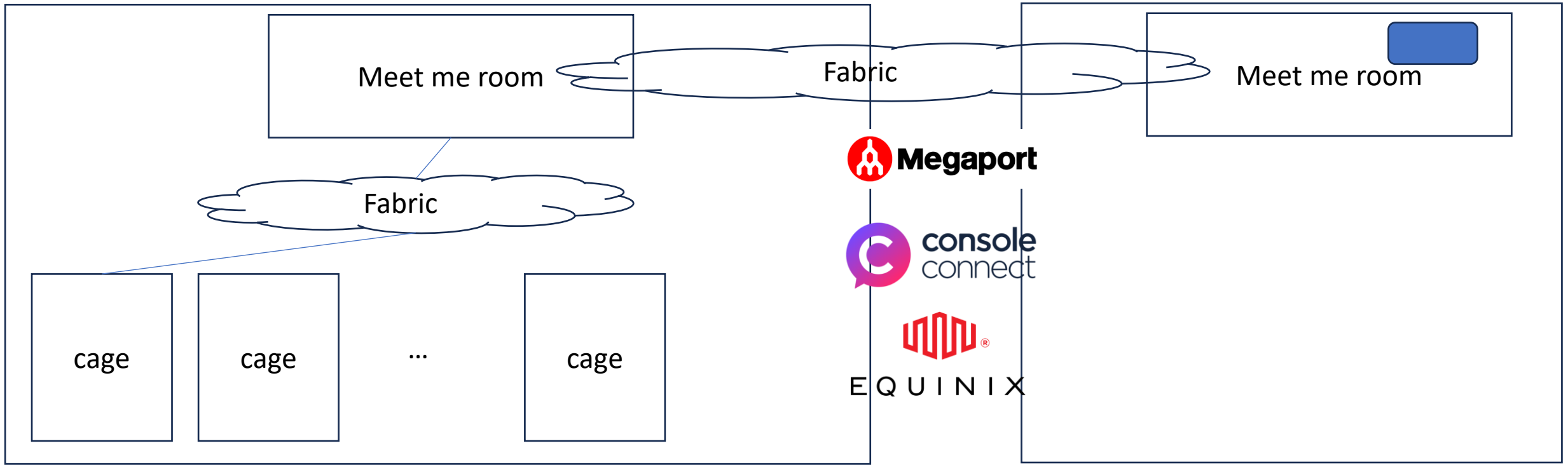


Could Use VPN



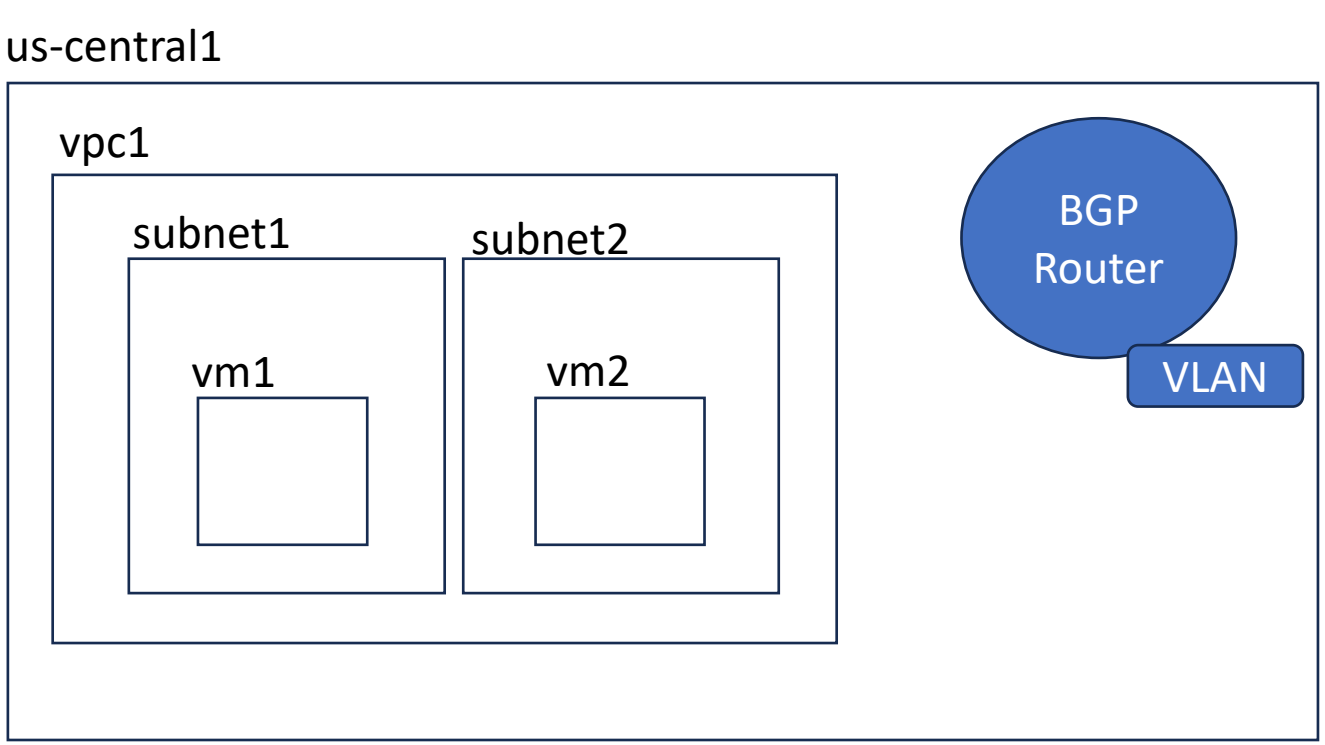
Recall: Cloud Direct Connect Providers

- Some provider's have a global network fabric and can 1) automate some of the connection process, 2) enable connections beyond where the cloud provider switch exists.

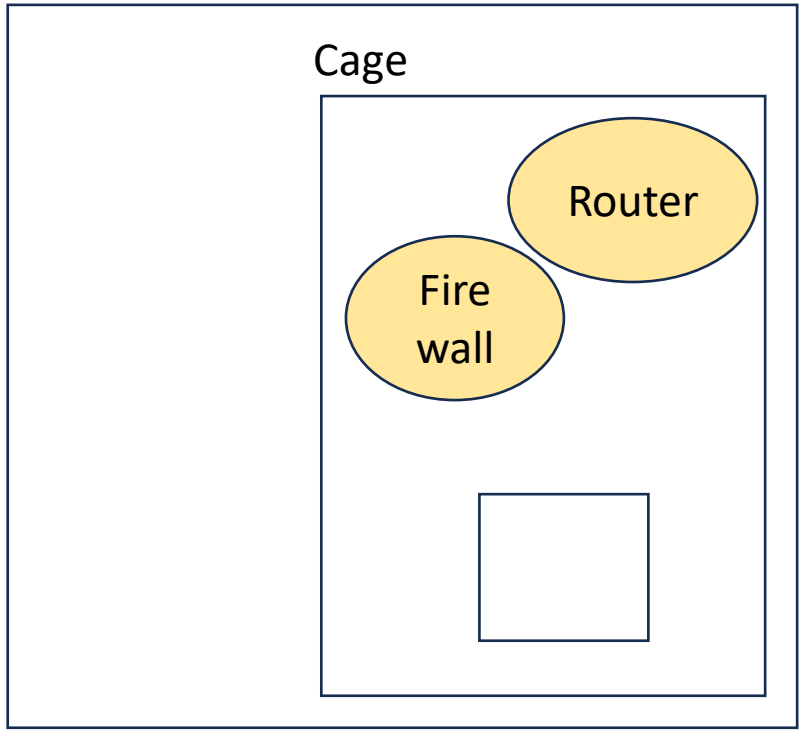


Recall: Direct Connect Example Setup

- After ordering the cross connect, and direct connect
- In cloud, need Router and VLAN attachment

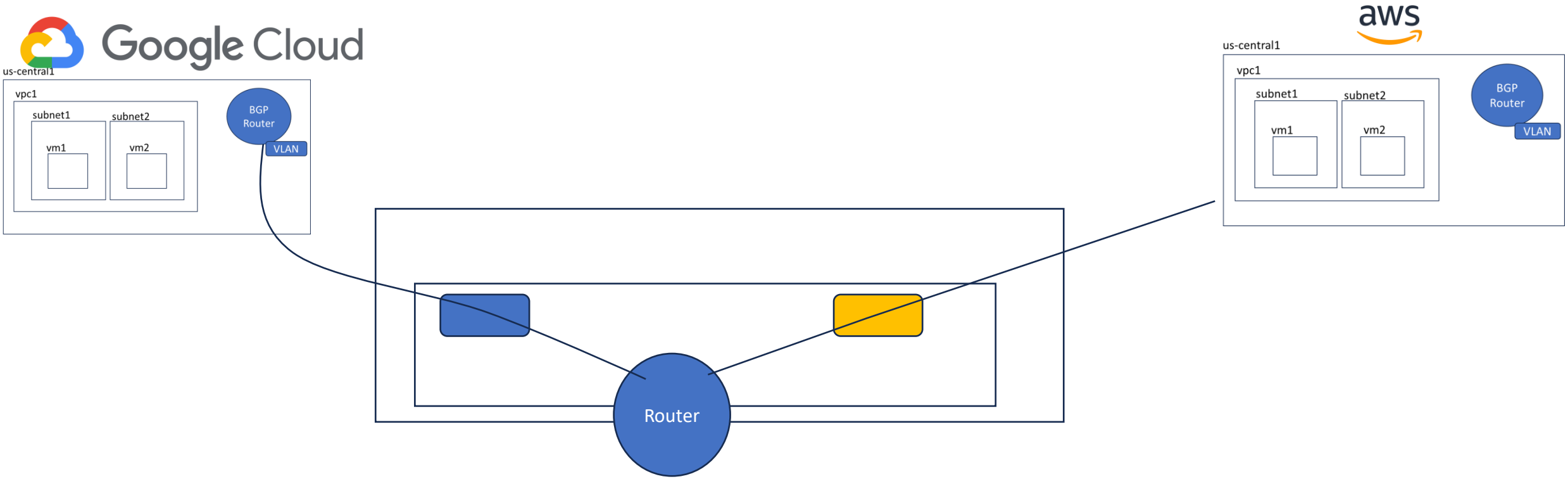


Co-location



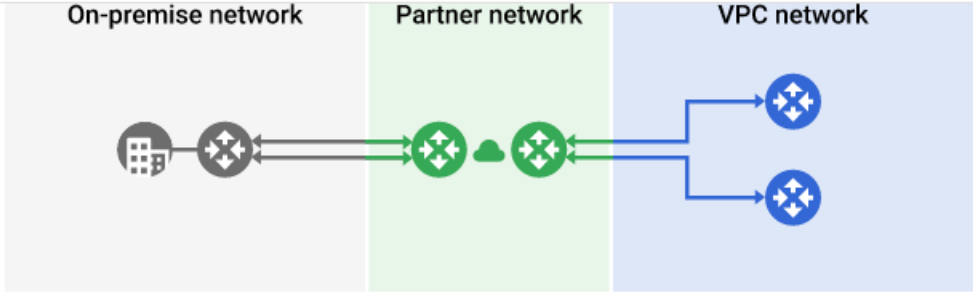
Cross Cloud Interconnect

- Effectively a direct connect to each cloud plus a virtual router
- Provider (Partner or Google) runs/operates the (virtual) router
- Can also work between different regions (over the fabric)



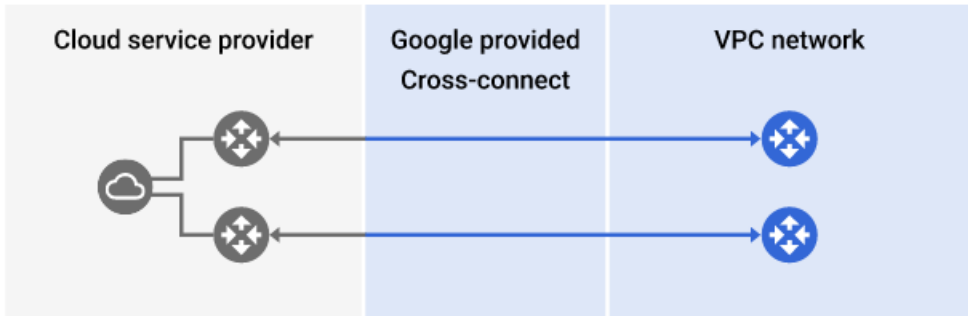
Network Connectivity Center
VPN
Interconnect
Cloud Routers

Add VLAN attachment



Cross-Cloud Interconnect connection

Connect your Google Cloud VPC network to your other cloud service provider network.
Learn more



Encrypt interconnect

Set up unencrypted Interconnect

Choose this option if you do not want to set up HA VPN over your Interconnect for encryption purposes

Set up HA VPN over Interconnect

Choose this option if you want HA VPN over your Interconnect connection. You must deploy on-premises IPsec devices to choose this option

CONTINUE CANCEL

Network Connectivity Center
VPN
Interconnect
Cloud Routers

Configure Cross-Cloud VLAN attachments

1. Select interconnects & redundancy
2. Create VLAN attachment for Interconnect A
3. Create VLAN attachment for Interconnect B
4. Configure BGP sessions
5. Configure your remote cloud resources

Select interconnects & redundancy

To create a VLAN attachment, first choose which Interconnect connection it will attach to.

Redundancy

Creating a redundant pair of VLANs is recommended to increase availability. If you don't need redundancy or an SLA, you can create a single VLAN attachment (and make it redundant later). Learn more

- (•) Create a redundant pair of VLAN attachments (recommended)
- ( ) Create a single VLAN (no redundancy)

Interconnects

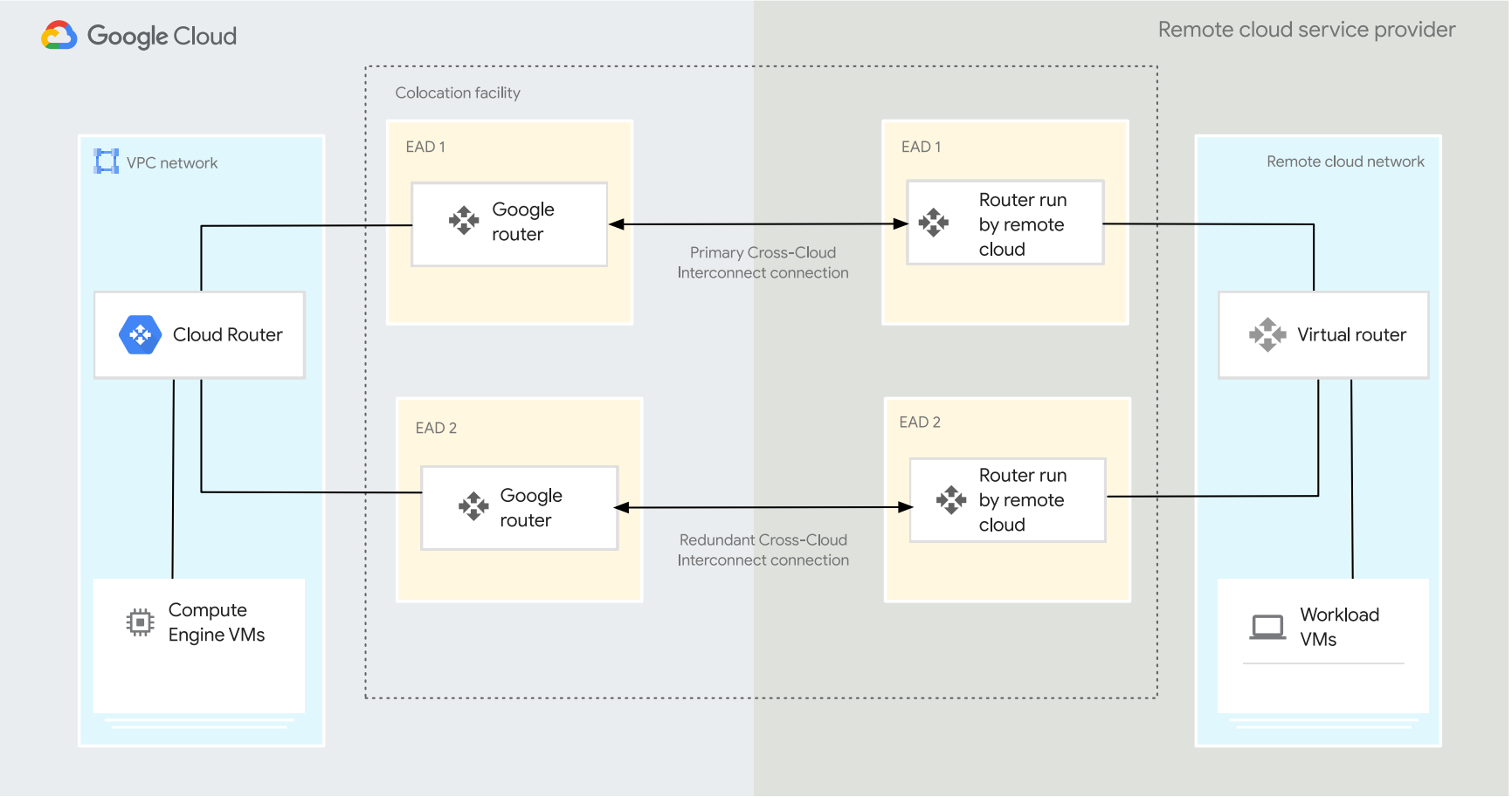
- (•) In this project
- ( ) In another project

⚠ You need to create an Interconnect connection in this project before you add VLAN attachments

Interconnect A *

Interconnect B *

End Result



https://cloud.google.com/network-connectivity/docs/interconnect/concepts/cci-overview



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